[2.2] Vanilla Policy Gradient

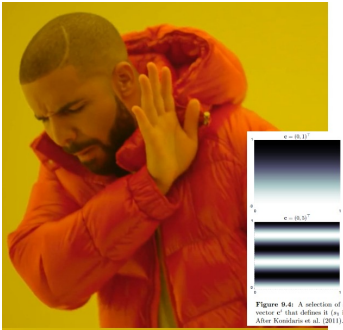
David Quarel

ILIAD Intensive

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Beyond Tabular Learning

- Tabular RL assumes that sweeping through all state-action pairs is tractable
- What about large/continuous state spaces?
  - State aggregation?
  - Parameterised policy $\pi_{\boldsymbol{\theta}}$, learn best $\boldsymbol{\theta}$?
  - Craft a heuristic by hand?
- In general, would like the agent to learn useful features for us
  - Something deep learning excels at!



9.5 Feature Construction for Linear Methods
- 9.5.1 Polynomials . . . . . .
- 9.5.2 Fourier Basis . . . . . .
- 9.5.3 Coarse Coding . . . . . .
- 9.5.4 Tile Coding . . . . . .
- 9.5.5 Radial Basis Functions . . . . . .

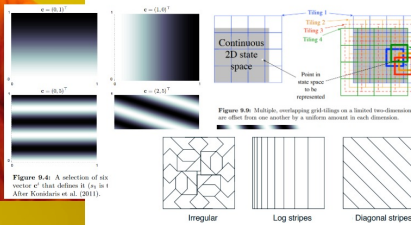
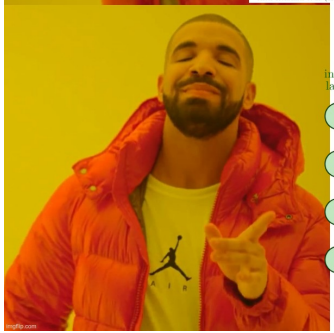
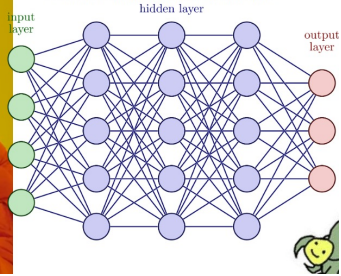


Figure 9.4: A selection of six vector $\mathbf{c}^i$ that defines it (x_1 in t
After Konidaris et al. (2011).

Figure 9.9: Multiple, overlapping grid-tilings on a limited two-dimension are offset from one another by a uniform amount in each dimension.

Figure 9.12: Tilings need not be grids. They can be arbitrarily shaped and non-u still in many cases being computationally efficient to compute.





Difficulties with using Neural Networks for RL

- Neural networks expect to be trained in a supervised learning fashion, with batches of data fed in, loss computed, and gradients backpropagated.
- We need a way to obtain a gradient somehow from trajectories, and then use this gradient to update the policy.
- Interaction with environment is *NOT* i.i.d, which supervised learning assumes, so we need to run many trajectories in parallel.
- Our grouth truth signal is if the policy obtains large returns during training.

VPG Framework

- **Action space** $\mathcal{A}$ is still discrete, and finite
- **State space** $\mathcal{S}$ is continuous (a subspace of $\mathbb{R}^n$), and each state is a vector $\mathbf{s} \in \mathcal{S}$.
- Environment is **episodic**, and agent always starts at/near **starting state** $\mathbf{s}_0$. We generate a full episode, and then afterwards learn. No discount ($\gamma = 1$).
- Define measure of performance (**joy**) as

$$J(\boldsymbol{\theta}) := V_{\pi_{\boldsymbol{\theta}}}(\mathbf{s}_0)$$

- Policy $\pi_{\boldsymbol{\theta}} \equiv \pi(\cdot|\mathbf{s}; \boldsymbol{\theta})$ is a conditional probability distribution $\pi_{\boldsymbol{\theta}} : \mathcal{S} \to \Delta\mathcal{A}$ over actions, mapping a state to a probability distribution over actions.
- Gradient $\nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}$ is well-defined.
- Choose $\boldsymbol{\theta}$ to maximize joy $J(\boldsymbol{\theta})$ via gradient *ascent*

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$$

Softmax policy

- In practice, $\pi_{\boldsymbol{\theta}} : \mathcal{S} \to \Delta\mathcal{A}$ is a neural network $\mathbf{h}$, mapping a state vector $\mathbf{s}$ to a vector of logits, one for each action $\mathcal{A} = \{a_1, \ldots, a_n\}$.

$$\mathbf{h}(\mathbf{s}; \boldsymbol{\theta}) = [h_1(\mathbf{s}; \boldsymbol{\theta}), \ldots, h_n(\mathbf{s}; \boldsymbol{\theta})]$$

  from which the action is sampled by softmax'ing the logits

$$\pi(a_i|\mathbf{s}; \boldsymbol{\theta}) := \frac{\exp h_i(\mathbf{s}; \boldsymbol{\theta})}{\sum_j \exp h_j(\mathbf{s}; \boldsymbol{\theta})}$$

- More principled than ϵ-greedy, as we get exploration for free, and exploratory actions are not selected at random, but with a bias towards actions believed to be good.

Policy Gradient Framework

- Assume episodic environment, episode length (horizon) T, no discount $\gamma = 1$.
- Assume fixed starting state s_0. Markov environment $\mu(s'|s, a)$.
- Define $J(\boldsymbol{\theta}) = V_{\pi_{\boldsymbol{\theta}}}(s_0) = \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}}}[G(\tau)]$.
- Let $\tau = s_0, a_0, r_1, s_1, \ldots, s_{T-1}, a_{T-1}, r_T, s_T$ denote a trajectory
- $G(\tau) = \sum_{t=1}^{T} r_t$ is the undiscounted return for trajectory τ.

$$\begin{aligned}\Pr(\tau|\boldsymbol{\theta}) &= \prod_{t=0}^{T-1} \pi_{\boldsymbol{\theta}}(a_t|s_t)\mu(s_{t+1}|s_t, a_t) \\ &= \pi_{\boldsymbol{\theta}}(a_0|s_0)\mu(s_1|s_0, a_0)\ldots\pi_{\boldsymbol{\theta}}(a_{T-1}|s_{T-1})\mu(s_T|s_{T-1}, a_{T-1})\end{aligned}$$

is the prob. of sampling trajectory τ from enviro. μ given params. $\boldsymbol{\theta}$.

Want to compute $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$.

Policy Gradient Framework

$$
\begin{aligned}
\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) &= \nabla_{\boldsymbol{\theta}} \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}}} [G(\tau)] \\
&= \nabla_{\boldsymbol{\theta}} \sum_{\tau} \Pr(\tau|\boldsymbol{\theta}) G(\tau) \\
&= \sum_{\tau} \nabla_{\boldsymbol{\theta}} \Pr(\tau|\boldsymbol{\theta}) G(\tau) \\
&= \sum_{\tau} \Pr(\tau|\boldsymbol{\theta}) \left(\nabla_{\boldsymbol{\theta}} \log \Pr(\tau|\boldsymbol{\theta}) G(\tau)\right) \qquad \text{as } \nabla \log f(x) = \frac{\nabla f(x)}{f(x)} \\
&= \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}}} [\nabla_{\boldsymbol{\theta}} \log \Pr(\tau|\boldsymbol{\theta}) G(\tau)]
\end{aligned}
$$

Problem: $\Pr(\tau|\boldsymbol{\theta})$ is unknown

Note that

$$\begin{aligned}
\nabla_{\boldsymbol{\theta}} \log \Pr(\tau|\boldsymbol{\theta}) &= \nabla_{\boldsymbol{\theta}} \log \prod_{t=0}^{T-1} \pi_{\boldsymbol{\theta}}(a_t|s_t)\mu(s_{t+1}|s_t, a_t) \\
&= \nabla_{\boldsymbol{\theta}} \sum_{t=0}^{T-1} \log \left(\pi_{\boldsymbol{\theta}}(a_t|s_t)\mu(s_{t+1}|s_t, a_t) \right) \\
&= \nabla_{\boldsymbol{\theta}} \sum_{t=0}^{T-1} \left(\log \pi_{\boldsymbol{\theta}}(a_t|s_t) + \log \mu(s_{t+1}|s_t, a_t) \right) \\
&= \nabla_{\boldsymbol{\theta}} \sum_{t=0}^{T-1} \log \pi_{\boldsymbol{\theta}}(a_t|s_t) + \cancel{\nabla_{\boldsymbol{\theta}} \sum_{t=0}^{T-1} \log \mu(s_{t+1}|s_t, a_t)} \\
&= \nabla_{\boldsymbol{\theta}} \sum_{t=0}^{T-1} \log \pi_{\boldsymbol{\theta}}(a_t|s_t)
\end{aligned}$$

REINFORCE

This gives the core equation for Vanilla Policy Gradient (VPG) or **REINFORCE**

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}}} \left[\sum_{t=0}^{T-1} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t|s_t) G(\tau) \right]$$

VPG generates a trajectory, learns, then discards it, generates another.

Clever trick 1: The future cannot affect the past

- In REINFORCE, each $\log \pi_{\boldsymbol{\theta}}(a_t|s_t)$ is multiplied by the full return $G(\tau)$.
- But a_t cannot influence rewards that happened **before** it.
- Can be shown that we can replace $G(\tau)$ with the **reward-to-go**:

$$G_t := \sum_{t'=t+1}^{T} r_{t'} = r_{t+1} + \ldots + r_T$$

- This gives a lower-variance, unbiased estimator.

$$\begin{aligned}\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) &= \mathbb{E}_{\tau\sim\pi_{\boldsymbol{\theta}}}\left[\sum_{t=0}^{T-1} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t|s_t) G(\tau)\right] \\ &= \mathbb{E}_{\tau\sim\pi_{\boldsymbol{\theta}}}\left[\sum_{t=0}^{T-1} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t|s_t) G_t\right]\end{aligned}$$

Algorithm 1 Vanilla Policy Gradient ($\gamma = 1$)

Input: Environment p, Number of episodes M

1: **for** episode $= 1$ to M **do**
2: Generate episode $s_0, a_0, r_1, s_1, \ldots, s_{T-1}, a_{T-1}, r_T s_T$
3: Define $G_t = \sum_{i=t}^{T-1} r_i$ for $0 \leq t \leq T-1$
4: Define gain $J(\boldsymbol{\theta}) = \sum_{t=0}^{T-1} G_t \log \pi_{\boldsymbol{\theta}}(a_t|s_t)$
5: Gradient **ascent** step $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \eta \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$
6: **end for**

VPG generates a trajectory, learns, then discards it, generates another.

Importance Sampling Theorem (IST)

For any two probability distributions p and q where $q(x) > 0$ whenever $p(x) > 0$:

$$\mathbb{E}_{x\sim p}[f(x)] = \mathbb{E}_{x\sim q}\left[\frac{p(x)}{q(x)}f(x)\right]$$

Proof:

$$\begin{aligned}
\mathbb{E}_{x\sim p}[f(x)] &= \sum_x p(x)f(x) \\
&= \sum_x \frac{p(x)}{q(x)}q(x)f(x) \\
&= \sum_x q(x)\frac{p(x)}{q(x)}f(x) \\
&= \mathbb{E}_{x\sim q}\left[\frac{p(x)}{q(x)}f(x)\right]
\end{aligned}$$

The ratio $\frac{p(x)}{q(x)}$ is called the **importance weight** or **importance ratio**.

VPG with Importance Sampling

Problem: VPG is on-policy: we must generate fresh trajectories for each policy update.
Solution: Use importance sampling to reuse old trajectories from $\pi_{\boldsymbol{\theta}_{old}}$.
The **exact** IS-corrected gradient requires the full trajectory probability ratio:

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}_{old}}} \left[\frac{\Pr(\tau|\boldsymbol{\theta})}{\Pr(\tau|\boldsymbol{\theta}_{old})} \sum_{t=0}^{T-1} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t|s_t) G_t \right]$$

$$= \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}_{old}}} \left[\prod_{t'=0}^{T-1} \frac{\pi_{\boldsymbol{\theta}}(a_{t'}|s_{t'})}{\pi_{\boldsymbol{\theta}_{old}}(a_{t'}|s_{t'})} \sum_{t=0}^{T-1} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t|s_t) G_t \right]$$

Problem: The product of ratios has *very high variance* for long trajectories.

VPG with Importance Sampling (cont.)

Common approximation: Use per-step importance ratios instead:

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) \approx \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}_{old}}} \left[\sum_{t=0}^{T-1} \frac{\pi_{\boldsymbol{\theta}}(a_t|s_t)}{\pi_{\boldsymbol{\theta}_{old}}(a_t|s_t)} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t|s_t) G_t \right]$$

$$= \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}_{old}}} \left[\sum_{t=0}^{T-1} \frac{\nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(a_t|s_t)}{\pi_{\boldsymbol{\theta}_{old}}(a_t|s_t)} G_t \right] \quad (\text{as } \pi \nabla \log \pi = \nabla \pi)$$

Why use the ratio at timestep t? For the term $\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t|s_t) G_t$:

- Action a_t was sampled from $\pi_{\boldsymbol{\theta}_{old}}(\cdot|s_t)$, but we evaluate $\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t|s_t)$
- Correct for this mismatch with $\frac{\pi_{\boldsymbol{\theta}}(a_t|s_t)}{\pi_{\boldsymbol{\theta}_{old}}(a_t|s_t)}$

What we ignore: State s_t and return G_t also depend on $\pi_{\boldsymbol{\theta}_{old}}$, but when $\pi_{\boldsymbol{\theta}} \approx \pi_{\boldsymbol{\theta}_{old}}$, these effects are small.

VPG with Importance Sampling Algorithm

Input: Environment, policy $\pi_{\boldsymbol{\theta}}$, batch size N, learning rate α
Algorithm: Repeat for each iteration:

1. Collect N episodes in parallel using current policy $\pi_{\boldsymbol{\theta}}$:
   - Generate episode $\tau^i = (s_0^i, a_0^i, r_1^i, s_1^i, \ldots, s_T^i)$
   - Store log probabilities: $\log \pi_{\text{old}}(a_t^i|s_t^i) = \log \pi_{\boldsymbol{\theta}}(a_t^i|s_t^i)$
   - Compute returns $G_t^i = \sum_{k=t+1}^{T} r_k^i$ for all t
   - Store $\{\tau^i, G_t^i, \log \pi_{\text{old}}(a_t^i|s_t^i)\}_{i=1}^{N}$.
2. Sample batch $\mathcal{B} = \{\tau^i, G_t^i, \log \pi_{\text{old}}(a_t^i|s_t^i)\}_{i=1}^{|\mathcal{B}|}$.
3. Compute importance sampling joy:

$$J(\boldsymbol{\theta}) = \frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \sum_{t=0}^{T-1} \frac{\pi_{\boldsymbol{\theta}}(a_t^i|s_t^i)}{\pi_{\text{old}}(a_t^i|s_t^i)} G_t^i$$

4. Gradient ascent: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$

Can reuse trajectories, or slice large rollouts into smaller batches.